JianghuSect 2 (JS2-v2.6.0) — Deep-Dive Feature & Logic Audit

System #03 (Part 1 of 2): Martial Arts & Techniques — Combat Moves (กระบวนท่า), Multi-Move Unlocking, AI Move Selection, Weapon-Move Matching & Spatial Execution

- **Target Codebase**: Js2 new (SIM.PATCH_VERSION = '2.6.0', SIM.SAVE_SCHEMA_VERSION = '2.6.0-living-chronicle-and-reporting')
- **Target Readers**: Autonomous Coding Models (**GPT** & **Gemini** Combat & Progression Engineers)
- **Core Simulation Philosophy**: Every martial art and combat move (กระบวนท่า) must emerge, unlock, and execute naturally through NPC training, sparring, weapon mastery, physiological/spiritual stats, and tactical combat AI — **zero dead catalog entries, zero fake tooltip numbers, zero player micromanagement**.
- **Modules Audited in Part 1**: js/skills.js, js/martial.js, js/combat.js, data/data_tech.js, data/data_martial.js, js/ai.js
- **Scope Split**:
  - **Part 1 (03A — This Document)**: Combat Move Generation (SK.profile / SK.moveProfile), Multi-Move Mastery Unlocking & Caching (SK.unlockedMoves, SK.list), AI Move Scoring & Selection (COMBAT.chooseSkill), Daily Training/Sparring Integration (A.practice, sparRound), Weapon-to-Move Compatibility (16 weapon classes), Spatial Range Caching (SK.resolveMoveSpatialSpec), and Cast/Channel/Charge Execution (COMBAT.useSkill).
  - **Part 2 (03B — Next Document)**: Composable Keyword Engine (COMPOSABLE_ENGINE), Multi-Tier Elemental Reactions (YIN_SHADOW vs DARK_YIN bug!), Battlefield Domains, Duet Skills, Passive Slots, Innate Subpath Arts, Mantras, Synergies & 3-Piece Resonance Sets.

---

0. Hard Invariants for AI Implementers

1. **Immutable Core Lock**: Never modify js/human_core.js (HC.FINGERPRINT === '155:2f9c9dd7').
2. **Script Load Order**: Keep exact 40 <script> tags in index.html (data_tech.js -> data_martial.js -> ... -> martial.js -> skills.js -> combat.js).

3. **Pure Read-Only Evaluation**: `SK.list`, `SK.moveProfile`, `SK.resolveMoveSpatialSpec`, `M.skillEfficiency`, and UI tooltip builders (`SK.summary`) must **NEVER** consume `W.rng` or mutate serialized state.
4. **80-NPC Performance Budget**: Median tick time must remain `<= 3.5 ms/tick` (all fixes below use `O(1)` version stamps rather than per-tick array rebuilds).

---

1. `[MOVE-CAT-01]` 54 `ART190_*` Techniques & Archetype Signature Finishers Are Invisible to Combat (`SK.profile === null` & Missing `SK.list` Wiring)

1.1 Exact Code Anchors

- `data/data_tech.js`: `JD.TECH.techniques` entries `ART190_XIAN_002..062`, `ART190_BODY_002..062`, `ART190_GOD_002..062`
- `js/skills.js`: `SK.profile` (lines 165–183), `SK.list` (lines 295–350), `SK.getArchetypeSignatureSkills` (lines 550–585)

1.2 Root-Cause Bug #1A — 54 Combat/Support Techniques in `JD.TECH.techniques` Return `SK.profile(t) === null`!

In `data_tech.js`, every 4th `ART190_*` entry inside `JD.TECH.techniques` (learned into `n.cult.techs`, **NOT** stored in `JD.TECH.mantras`, and having **no `passive` object and no `mantraBonus`**) was tagged with `cat: "QI_MANTRA"`, `"BODY_MANTRA"`, or `"FAITH_MANTRA"` despite being an active martial art with `role: "COMBAT_DEFENSE"` or `"HEAL_SUPPORT_TRAINING"` and 3 named `moves`:

- Examples of affected active martial arts:
  - `ART190_XIAN_002` (เคล็ดกระบี่วายุคราม ตัดคลื่น), `ART190_XIAN_018` (เพลงดาบเปลวสุริยันผ่าผา), `ART190_XIAN_022` (เพลงขลุ่ยหยกเรียกพายุ), `ART190_XIAN_026` (เพลงกระบี่สะท้อนเงาวารี), `ART190_XIAN_030` (ฝ่ามือหมอกพิษบุปผาร่วง)
  - `ART190_BODY_002` (เพลงทวนมังกรโลหิตกวาดทัพ), `ART190_BODY_014` (ท่าทุ่มมังกรจับพยัคฆ์), `ART190_BODY_026` (เพลงทวนสั้นทะลวงจุดตาย), `ART190_BODY_030` (เพลงดาบใหญ่หัวตัดสะบั้นค่าย), `ART190_BODY_034` (กรงเล็บโลหิตเดือดฉีกวิญญาณ)
  - `ART190_GOD_002` (ฝ่ามือเทพเมตตาปัดเป่าเคราะห์), `ART190_GOD_006` (เพลงทวนทองคำพิทักษ์วิหาร), `ART190_GOD_034` (วิชาค้อนเทวทัณฑ์ทุบทำลายแกนมาร)
  - Plus 12 `HEAL_SUPPORT_TRAINING` entries (`053, 056, 059, 062` across `XIAN`/`BODY`/`GOD`, e.g., `ART190_XIAN_053` เคล็ดเข็มทองเปิดชีพจรขับพิษ, `ART190_GOD_053` มนตราน้ำอมฤตเมตตาชุบชูใจ)!

Now trace `SK.profile(t)` (`js/skills.js:167–183`):

```javascript
else if (t.cat === 'AUXILIARY_ART' && t.role === 'HEAL_SUPPORT_TRAINING') { ... }
else if (t.cat === 'COMBAT_TECHNIQUE') { ... }
else if (Array.isArray(t.moveDefs) && t.moveDefs.length) { ... }
else return (cache[t.id] = null); // วิชาอาชีพ/ไม่ใช่การรบ
```

Because `t.cat` is `'QI_MANTRA' | 'BODY_MANTRA' | 'FAITH_MANTRA'` (not `'COMBAT_TECHNIQUE'` and not `'AUXILIARY_ART'`) and `t.moveDefs` is undefined, **`SK.profile(t)` returns `null` for all 54 techniques**!

- **Impact**: Disciples spend hundreds of Contribution Points (`CP`) and study hours learning *"เพลงทวนมังกรโลหิตกวาดทัพ"* or *"เพลงดาบใหญ่หัวตัดสะบั้นค่าย"*, yet they gain **0 passive stats, 0 mantra bonuses, and 0 combat moves**! Even `SK.TECH_TO_T6_VFX` (`skills.js:760+`) mapped VFX onto `ART190_XIAN_030`, `ART190_BODY_030`, and `ART190_GOD_030` without realizing those techniques can never fire in combat!

1.3 Root-Cause Bug #1B — All 32 Core Mantras in `JD.TECH.mantras` Have Named `moves` That Never Manifest, and `SK.getArchetypeSignatureSkills` (`T4/T6` Finishers) Never Enter `SK.list`

1. **32 Non-Legendary Mantras (`data_tech.js:505–570`)**: Every entry in `JD.TECH.mantras` (from `MANTRA_BASIC_BREATH` to Tier-7 `MANTRA_PRIMORDIAL_CHAOS_QI` / `MANTRA_PRIMORDIAL_DRAGON_BODY`) defines 2 named moves (`moves: ["เบิกฟ้าแยกปฐพี", "ปราณโกลาหลกลืนกินหมื่นวิถี"]`), yet `SK.list` (`skills.js:321`) only includes `n.cult.mantra` if `Array.isArray(mTech.moveDefs)` (which only the 10 Legendary Mantras in `data_martial.js` have!).
2. **Orphaned Tier-4 & Tier-6 Subpath Archetype Signature Finishers (`skills.js:550–585`)**: `SK.getArchetypeSignatureSkills(archId, tier)` creates `ARCH_<ID>_T4` and `ARCH_<ID>_T6_FINISHER`, **but `SK.list(n)` (`skills.js:295–350`) never appends `n.archetypeMastery.unlockedSkills` (or calls `SK.getArchetypeSignatureSkills`) into `out`**, and those objects also lack `costPct`, `hits`, `elemTh`, `range`, and `summary`!

1.4 Remediation Blueprint

1. In `SK.profile(t)` (`js/skills.js:173–183`):

- Match `t.role === 'HEAL_SUPPORT_TRAINING'` regardless of whether `t.cat` is `'AUXILIARY_ART'` or `/*_MANTRA$/`.
- Match `(t.cat === 'COMBAT_TECHNIQUE' || t.role === 'COMBAT_DEFENSE' || t.role === 'COMBAT_OFFENSE' || (/_MANTRA$/.test(t.cat || '') && Array.isArray(t.moves) && t.moves.length > 0 && !t.passive))` so all 54 `ART190_*` techniques and equipped `n.cult.mantra` arts generate real, balanced combat moves!
- Also run `inferWeaponTypesFromName(t)` in `data_martial.js` on `role === 'COMBAT_DEFENSE' | 'COMBAT_OFFENSE'` entries so `"เพลงทวนมังกรโลหิต"` gets `weaponTypes: ['SPEAR']` and `"เพลงดาบเปลวสุริยัน"` gets `weaponTypes: ['BLADE']`.

2. In `SK.list(n)`:
   - Allow `n.cult.mantra` to contribute its `SK.profile(mTech)` (at a balanced `1.15x` cd so it complements rather than crowds out combat manuals) even when `mTech.moveDefs` is absent.
   - If `n.subpathArchetypeMastery` has `tier >= 4`, enrich `SK.getArchetypeSignatureSkills` through `SK.resolveMoveSpatialSpec` + `SK.summary` and append them to `SK.list(n)`.

---

2. `[MOVE-CACHE-02]` `SK.list` Cache (`_skListCache`) Blind to Mastery (`c.tm`) & `effSt`, Locking Multi-Move Unlocks + Save-State `c.tm` Key Bloat

2.1 Exact Code Anchors

- `js/skills.js`: `SK.unlockedStage` (lines 232–246), `SK.unlockedMoves` (lines 256–268), `SK.gainMastery` (lines 277–293), `SK.list` (lines 295–350)
- `js/combat.js`: `useSkill` mastery increment (lines 805–813)

2.2 Root-Cause Chain Reaction

1. **Stale `_skListCache` Blocks Move 2 / Move 3 Unlocks (`skills.js:304–306`)**:

```javascript
if (slc && slc.path === c.path && slc.sub === c.sub && slc.st === c.stage &&
slc.mantra === c.mantra && slc.tl === tl && slc.tSig === tSig && slc.eqKey ===
eqKey && slc.ver === (n._passVer || 0)) {
  return slc.out;
```

```
}
```

- Hundreds of multi-move techniques (`v2.0.34`, `v2.0.0`, `v2.4.0`) define `moveDefs: [{ id: '..._M1', reqMastery: 0 }, { id: '..._M2', reqMastery: 20 }, { id: '..._M3', reqMastery: 45 }]` **without a `t.stages` array**.
- When `SK.gainMastery` runs (`line 279`), `SK.unlockedStage` returns `1` before and `1` after (`if (!Array.isArray(t.stages) || !t.stages.length) return 1`), so `newStage > prevStage` is **never `true`** and `n._passVer` is **never bumped**!
- Meanwhile `_skListCache` does **not** track `n._tmVer` (mastery changes) and checks raw `c.stage` instead of `effSt = CULT.effectiveTechStage(n)`!
- **Result**: Even after an NPC accumulates `50+` mastery points on a `stages`-less multi-move manual, `_skListCache` keeps returning `[Move 1]` only!

2. **Duplicate Per-Move Mastery Bloat (`combat.js:813`)**:
   - `SK.gainMastery(W, a, techId, masGain)` already increments `a.cult.tm[techId]`.
   - Line 813 then executes `if (p.id !== techId) { a.cult.tm[p.id] = (a.cult.tm[p.id] || 0) + masGain; }` — creating thousands of orphaned `tm[moveId]` keys across 80 NPCs that `SK.unlockedMoves` never reads and `SK.masteryBase` ignores because `c.tm[techId]` is truthy first!

2.3 Remediation Blueprint

- In `SK.unlockedMoves(n, t)`, check whether crossing `m.reqMastery` unlocked a new move count in `SK.gainMastery` (compare `prevMoveCount` vs `newMoveCount`), bump `M.invalidateMartialCaches(n)` whenever `newStage > prevStage || newMoveCount > prevMoveCount`, and log `'insight'` when a new move (`m.nameTh`) is comprehended!
- Include `effSt` in `_skListCache` (`slc.effSt === effSt`), and remove the redundant `a.cult.tm[p.id]` write in `combat.js:813` (plus strip orphaned move-ID keys from `c.tm` on deserialize).

---

3. [MOVE-DATA-03] SK.moveProfile Missing Archetype Payloads (channel/summon/dot) + Copy-Paste Fake Multipliers & Wrong Damage Types in data_martial.js

3.1 Exact Code Anchors

- js/skills.js: SK.moveProfile (lines 106–163) vs SK.profile (lines 184–226)
- data/data_martial.js: MP_BOW_* (lines 35–75) and EXP_V204.newInnateTechniques (lines 700–1150)

3.2 Root-Cause Defects

1. **SK.moveProfile Drops Archetype Payloads When m Only Specifies arch**:
   - In SK.moveProfile (lines 131–138):
     - dot: m.dot || 0 (ignores A.dot = 0.8 on arch === 'dot'!)
     - channel: m.channel || null (leaves prof.channel === null on arch === 'channel' moves like INNATE_ASTRAL_ARRAY_CORE_M2 and INNATE_SOUND_ZITHER_CORE_M2, so useSkill skips efficiency/synergy/element scaling on p.channel and falls back to flat unscaled { tickM: 0.5, finishM: 1.0 }, while SK.summary hides all damage text!)
     - summon: m.summon || null (leaves prof.summon === null on arch === 'summon' moves like INNATE_PHANTOM_SOUL_CORE_M2, INNATE_TOTEM_BEAST_GOD_CORE_M1, INNATE_NETHER_GHOST_KING_CORE_M1, INNATE_DREAM_ILLUSION_WEAVER_CORE_M2!)
2. **Non-Attacking Moves (guard, shield, heal, support, buff, summon, evade) Authored With M: 1.42 / 1.95 & pen: 18 / 28 (data_martial.js:700–1050)**:
   - In EXP_V204.newInnateTechniques, every generated _M1 / _M2 move was stamped with M: 1.42 or M: 1.95, hits: 1, pen: 18/28, critAdd: 10 — **even when arch is 'guard', 'heal', 'support', or 'summon'!**
   - In COMBAT.useSkill (combat.js:880–960), guard, shield, evade, heal, support, buff, and summon **all return before the attack loop**!
   - Consequently, SK.summary lies to the player ("ดาเมจ ×1.95 · เจาะเกราะ +28%" on a pure heal/guard/summon move!), and chooseSkill evaluates bogus dmgEq before overwriting sc.
3. **Physical Bow Assassins (SHADOW_ARCHER, IRON_BOW_SENTINEL, STAR_ARCHER) Scaling Off Magic (type: 'm') & SK.profile Defaulting 34 Grand Dao Paths to 88% Magic**:

- `MP_BOW_SHADOW_SNIPE` (`M1/M2`) and `INNATE_SHADOW_ARCHER_CORE` (`M1/M2`) belong to `SHADOW_ARCHER` (`DEMON`, `+14% atk`, `0% matk`), yet their moves have `type: 'm'`!
- In `SK.profile` (`skills.js:189–198`), `type` only checks 9 old paths and sends **all 34 other paths** (including `SHADOW_SAMURAI`, `NINPO_SHADOW`, `ARENA_GLORY`, `DEVOURING_FLESH`, `PRIMORDIAL_SHIFTER`, `GRAVITY_DAO`) to `rnd(h, 3) < 0.12 ? 'p' : 'm'` (**88% Magic**)!

3.3 Remediation Blueprint

- In `SK.moveProfile`:
  - Synthesize canonical `prof.channel`, `prof.summon`, and `prof.dot` defaults (matching `SK.profile`'s tier-scaled formulas and deriving `tickM`/`finishM` from `m.M` when `m.M > 0`!) whenever `arch` is `'channel'`, `'summon'`, or `'dot'` and the explicit object is omitted.
  - Force `prof.M = 0; prof.hits = 0; prof.pen = 0; prof.critAdd = 0;` when `['guard', 'shield', 'heal', 'support', 'buff', 'summon', 'evade'].includes(arch)` (unless `arch === 'command' || arch === 'harmonize'` which legitimately hybrid-strike!).
- Fix `MP_BOW_SHADOW_SNIPE` and `INNATE_SHADOW_ARCHER_CORE` to `type: 'p'`, and update `SK.profile`'s `type` resolver to read `JD.CULT.paths[t.path].statWeights` (`atk >= matk ? 'p' : 'm'`) across all 43 paths!

---

4. `[MOVE-AI-04] chooseSkill` Threshold Trap (`0.62` Factor vs `bs = 1.15`) Starving Starter Moves + `sc =` Overwrites Erasing Tier Scaling

4.1 Exact Code Anchors

- `js/combat.js`: `chooseSkill` (lines 614–716)

4.2 Mathematical Proof of Two Severe Combat AI Bugs

Bug 4A — Starter & Physical-on-Hybrid Moves Score `< 1.15` (`bs`), So NPCs Spam Weak Auto-Attacks Instead of Their Martial Arts!

Inspect lines 625 and 641 of `js/combat.js`:

```JavaScript
let best = null, bs = 1.15;
...
const basicStat = Math.max(sa.atk, sa.matk);
const stat = p.type === 'p' ? sa.atk : sa.matk, rel = stat / Math.max(1,
basicStat), mas = a.foe ? 1 : SK.mastery(a, p);
if (p.M) {
  const dmgEq = p.M * elemDmgMult * (p.hits || 1) * rel * mas * ...;
  sc = dmgEq * (4 / Math.max(3, effWind)) * 0.62;
  sc += aggr * 0.5;
}
```

Now plug in real numbers:

1. **Case 1 — `BODY` Disciple with Starter Technique `ART190_BODY_001` Move 1 (พยัคฆ์คำรามสั่นภูผา: `M = 1.30, wind = 3, type = 'p'`)**:
   - Even with `rel = 1.0` and `elemDmgMult = 1.0`:
   - `sc = 1.30 * (4 / 3) * 0.62 = 1.075`!
   - **`1.075 < 1.15` (`bs`)!** Even though this move deals **1.30x damage in 3 windup ticks** (`0.433 M/tick`) vs a basic attack's **1.00x damage in 4 windup ticks** (`0.250 M/tick` — **+73% stronger!**), `chooseSkill` **rejects it and returns `null`**, forcing the disciple to throw nameless basic punches!
2. **Case 2 — `XIAN` Sword Immortal (`SWORD_IMMORTAL`) with `ART190_XIAN_001` (เพลงกระบี่พิรุณหยกเจ็ดดาว) or `INNATE_SWORD_IMMORTAL_HEART` (`type = 'p'`)**:
   - `XIAN` grants `+14% matk` and spiritual foundation growth, so `sa.matk > sa.atk` (`rel = sa.atk / sa.matk ≈ 0.68`).
   - `sc = 1.25 * 0.68 * 1.15 * (4 / 3) * 0.62 = 0.808 < 1.15`!
   - **Domino Effect**: Because `chooseSkill` returns `null`, the Sword Immortal **never casts Move 1**, which means `useSkill` never runs `SK.gainMastery`, which means **Move 2 (เจ็ดดาวเรียงร้อย) and Move 3 (คมหยกสะบั้นเมฆา) NEVER UNLOCK**!

Why did this happen? Because basic attack baseline on the `* 0.62` scale is **`1.0 * (4 / 4) * 0.62 = 0.62`**, NOT `1.0`! Setting `bs = 1.15` accidentally demanded **`1.15 / 0.62 = 1.85x` basic-attack DPS** just to cast any offensive skill!

Bug 4B — `sc = ...` Overwrites Erase `dmgEq`, Tier, Element, Synergy & Weapon Modifiers on `control`, `counter`, and `channel` (`lines 643, 647, 662`)!

- Line 641 computes `sc` from `p.M`, `elemDmgMult`, `rel`, `mas`, and `effWind`.
- Immediately afterward:
  - Line 643: `if (p.arch === 'control') sc = (ST.has(...) ? 0.2 : 1.7) * ...;` (**Overwrites `sc`!**)

    - Line 647: `if (p.arch === 'counter') sc = (ST.has(...) ? 0 : 1.1 + caut) * ...;` (**Overwrites `sc`!**)
    - Line 662: `if (p.arch === 'channel') sc = (ctx.channeling ? 0 : 2.2) + ...;` (**Overwrites `sc`!**)
- **Result**: A Tier-1 mismatched-weapon `control` or `channel` move (`M = 0.7, mas = 0.55`) gets the **exact same base score** as a Tier-7 Legendary `control` or `channel` move (`M = 2.15, mas = 1.45`, e.g., *Dugu Nine Swords Move 5*)!

4.3 Remediation Blueprint

1. Compare offensive moves against the **actual basic-attack baseline** (`basicBaseline = 0.72` on the `0.62` scale, i.e., `bs = 0.78`), so any move whose tempo-adjusted output beats a basic auto-attack by `>= 10%` is cast!
2. For Sword/Hybrid paths (`SWORD_IMMORTAL`, `MIND_SWORD_TELEKINESIS`, `CALLIGRAPHY_PAINTING_SAGE`, `ASTRAL_ARRAY`) wielding matched weapons, let sword-qi techniques (`weaponTypes` includes `'SWORD'`) blend `max(sa.atk, sa.matk * 0.88)` (or let `rel` compare against basic attack's actual stat!).
3. Change `control`, `counter`, and `channel` in `chooseSkill` from destructive `sc = ...` to **additive / multiplicative blending with `dmgEq`** (e.g., for `control`: `sc = sc * 0.55 + ctrlUtility`; for `channel`: calculate `dmgEq` from `(p.channel.tickM * (ticks - 1) + p.channel.finishM)` and add `+ 1.1` when `!ctx.channeling`; for `counter`: scale utility by `(p.M || 1) * mas * 0.65`)!

---

5. `[MOVE-PRAC-05]` Contextual Disconnect: Years of Daily `practice` & Friendly `spar` Grant Zero Technique Mastery (`c.tm`)

5.1 Exact Code Anchors

- `js/ai.js`: `A.practice.tick` (lines 345–365)
- `js/combat.js`: `C.sparRound` (lines 20–30)

5.2 Contextual & Emergent-World Absurdity

In Wuxia/Xianxia lore, disciples unlock the deeper forms (ขั้นที่ `2..6`) of their sect's swordplay by **practicing forms at the martial ground (`practice`) and crossing hands with fellow disciples (`spar`)**. Currently:

- `A.practice.tick` trains body physiology, weapon skill (`h.skills.SWORD`), cultivation XP, and subpath `c.track` — **it NEVER increments `n.cult.tm[techId]`**!

- `C.sparRound` rolls `CULT.power(a) * (0.7 + rng * 0.6)` with generic `'⚔'` text — **it never selects a learned move from `SK.list(a)` and never increments `n.cult.tm[techId]`!**
- **Result**: A peaceful inner disciple who trains 6 hours a day for 50 years has `0` mastery in every martial art they studied from the Scripture Pavilion until a lethal bandit raid happens!

5.3 Remediation Blueprint

1. In `A.practice.done` (or every 30 ticks in `A.practice.tick`): pick a compatible active combat/movement/support technique in `n.cult.techs` (prioritizing techniques that still have locked higher `stages` / `moveDefs`!) and call `SK.gainMastery(W, n, techId, 0.35 * (1 + comprehension / 140))`, displaying the practiced move name (`m.nameTh`) in `n.fx`!
2. In `C.sparRound`: pick an actual unlocked move from `SK.list(a)` and `SK.list(b)` (weighted by weapon match and counter-tactics via `C.resolveWeaponTactics`), show the real move name in `a.fx` / `b.fx`, and grant `SK.gainMastery(W, x, move.tech, 0.18)` per exchange!

---

6. `[MOVE-WPN-06]` Weapon-to-Move Classification Bugs Across the 16 Weapon Classes (`data_martial.js`, `data_tech.js`, `skills.js`)

6.1 Exact Code Anchors

- `data/data_martial.js`: `D.types` (line 4), `D.classify` (lines 18–21), `inferWeaponTypesFromName` (lines 503–518)
- `data/data_tech.js`: `ART190_*` explicit `weaponTypes: ["FOCUS"]` / `["FIST"]`
- `js/skills.js`: `SK.inferFoeWeaponType` (lines 466–482)

6.2 Smoking-Gun Weapon Mismatches (Wielding the Exact Named Weapon Gives a `0.65x` Penalty!)

Because `ART190_*` and `data_martial.js` were authored when only 7–10 weapon classes existed, techniques explicitly named after **Fan, Zither, Whip, Brush, Needle, Abacus, Rosary, and Claw** still have `weaponTypes: ["FOCUS"]` or `["FIST"]`:

- `ART190_XIAN_007` (เพลง**พัด**หยกพายุหิมะพัดพา) & `ART190_GOD_028` (วิชา**พัด**ขนนกยูงเทวะปัดเป่า) -> `weaponTypes: ["FOCUS"]`! Wielding a **`FAN`** suffers **`0.65x` penalty (`ADAPTED_WEAPON`)**!

- ART190_XIAN_008 (วิชา**พิณ**เจ็ดสายสะกดวิญญาณ) -> weaponTypes: ["FOCUS"]! Wielding a **ZITHER** suffers **0.65x penalty**!
- ART190_XIAN_011 (เพลง**แส้**เถาวัลย์วิญญาณมรกต) -> weaponTypes: ["FOCUS"]! Wielding a **WHIP** suffers **0.65x penalty**!
- ART190_XIAN_012 (วิชา**ลูกคิดทองคำ**คำนวณจุดตาย) -> weaponTypes: ["FOCUS"]! Wielding an **ABACUS** suffers **0.65x penalty**!
- ART190_XIAN_021 (วิชา**เข็มเงิน**เมฆาดารา) & ART_DEMON_VENOM_NEEDLE_T1 (ดัชนี**เข็มพิษ**แมงป่องหยก) -> weaponTypes: ["FOCUS"]! Wielding a **NEEDLE** suffers **0.65x penalty**!
- ART190_XIAN_027 (วิชา**พู่กัน**ชาดจารึกอักขระศึก) & ART_SOUL_INK_DOT_T1 -> weaponTypes: ["FOCUS"]! Wielding a **BRUSH** suffers **0.65x penalty**!
- ART190_BODY_005 (กรงเล็บหมีเหมันต์), ART190_BODY_020 (กรงเล็บอินทรีเหล็ก), ART_DEMON_BLOOD_SIPHON_CLAW_T1, TECH_ANCESTRAL_DRAGON_CLAW -> weaponTypes: ["FIST"]! Wielding a **CLAW** suffers **0.65x penalty**!
- Also in D.classify(slot, name) (data_martial.js:18): /กรงเล็บ/ is classified as 'FIST' instead of 'CLAW', and FAN, NEEDLE, ABACUS, ASTROLABE, ROSARY fall through to 'FOCUS'!
- Also in M.weaponCompatibility (martial.js:218): severe ranged/melee mismatch checks only 'BOW' — firing a ZITHER or ASTROLABE move with a heavy BLADE or SWORD move while holding a BOW is inconsistent across the newer ranged weapons.

6.3 Remediation Blueprint

- Run a deterministic catalog-normalization pass at the end of data_martial.js (after all 16 weapon classes are loaded) that expands t.weaponTypes when t.nameTh explicitly names:
  - /พัด/ -> prepend 'FAN'; /พิณ|ขลุ่ย|สังคีต/ -> prepend 'ZITHER'; /แส้|โซ่|ไหมฟ้า/ -> prepend 'WHIP'; /พู่กัน|อักษร/ -> prepend 'BRUSH'; /เข็ม/ -> prepend 'NEEDLE'; /ลูกคิด/ -> prepend 'ABACUS'; /ประคำ/ -> prepend 'ROSARY'; /จานดารา|หลัวผาน|ดาราพยากรณ์/ -> prepend 'ASTROLABE'; /กรงเล็บ/ -> include 'CLAW' alongside 'FIST'! (Keep 'FOCUS'/'FIST' as secondary valid entries where backward compatibility tests expect them!)
- Update D.classify and SK.inferFoeWeaponType to recognize all 16 weapon classes.

---

7. [MOVE-SPATIAL-07] Pre-Cached prof.range (u = null) Kills Dynamic Stance & Weapon Reach + Over-Broad Regex Range Overrides

7.1 Exact Code Anchors

- js/skills.js: SK.moveProfile (line 161), SK.profile (line 225), SK.resolveMoveSpatialSpec (lines 400–460)
- js/combat.js: evaluateTacticalEngagement (line 532), chooseSkill (line 628), useSkill (line 842)

7.2 Root-Cause Bugs

1. **Static prof.range Short-Circuits SK.resolveMoveSpatialSpec(p, u)**:
   - SK.moveProfile (line 161) and SK.profile (line 225) assign: prof.range = m.range || t.range || SK.resolveMoveSpatialSpec(prof, null);
   - Then combat.js writes: const mSpec = p.range || SK.resolveMoveSpatialSpec(p, a);
   - Because p.range is **already truthy** (cached with u = null!), SK.resolveMoveSpatialSpec(p, a) is **NEVER CALLED WITH THE WIELDER a**!
   - **Dead Features Caused by This**:
     - THUNDER_STEP_STANCE / PUPPET_OVERDRIVE_MODE (+0.55 range) and VAJRA_BASTION_STANCE (-0.25 range) in skills.js:441–447 **never take effect**!
     - Weapon-unrestricted moves (wTypes = []) always resolve with primaryWpn = 'FIST' (maxR = 1.25) even when cast by a SPEAR (2.50) or BOW (3.50) wielder!
     - M.applyEquipmentSkillModifiers with mod.makeAoe = true (martial.js:548) leaves p.range stuck at shape: 'SINGLE', radius: 0!
2. **Regex Matching parentNameTh Turns Melee Punches Into 3.6-Tile Ranged Spells (skills.js:405, 420)**:
   - nameStr concatenates p.nameTh + ' ' + p.parentNameTh.
   - ART190_BODY_018 (หมัดภูผาถล่มทลายตา**ค่ายกล**) matches /ค่ายกล/ -> a FIST punch gets maxR = 3.60!
   - พายุทวนสังหารหมู่ (melee spear spin) matches /พายุ/ -> gets maxR = 3.60!

7.3 Remediation Blueprint

- Store explicit authored range in `prof.explicitRange = m.range || t.range || null` and default preview range in `prof.baseRange = SK.resolveMoveSpatialSpec(prof, null)`.
- In `combat.js`, always call `SK.resolveMoveSpatialSpec(p, a)` (which can memoize in `O(1)` by `(p.id, equippedWeaponType, activeStance, p.aoe)`!).
- Restrict `TECHNIQUE_OVERRIDE_MID` (`/ค่ายกล|อาคม|พายุ/`) so it does **not** override moves whose `weaponTypes` are strictly melee (`FIST`, `BLADE`, `SPEAR`, `STAFF`, `CLAW`) unless `p.type === 'm'` and the move name itself denotes a ranged wave/projectile!

---

8. `[MOVE-EXEC-08]` Execution Pipeline Bugs (`TECH_GCD` Asymmetry, `startChargedSkill` Scaling Loss, `slotSig` Mismatch & Zero-Qi Free Ultimates)

8.1 Four Concrete Execution Bugs & Exact Fixes

Bug ID	Exact Anchor	What Goes Wrong	Exact Fix
8A. Move 1 Skips `TECH_GCD`	`skills.js:112` & `combat.js:799`	`SK.moveProfile` sets `prof.id = isPrimary ? t.id : m.id`. In `useSkill`, `if (p.tech && p.tech !== p.id)` is **`false`** for Move 1, so Move 1 never triggers `'TECH_GCD:' + p.tech`, allowing instant Move 1 -> Move 2 chaining with zero shared windup GCD!	Check `const parentTech = CULT.tech(p.tech); if (parentTech && Array.isArray(parentTech.moveDefs) && parentTech.moveDefs.length > 1)` to set `'TECH_GCD:' + p.tech` on **all** moves of a multi-move manual (including Move 1)!
8B. `startChargedSkill` Drops	`combat.js:856, 1098, 1685`	`startChargedSkill` sets `tickM: 0.35` (fires an immediate	Set `tickM: 0` on `isChargedSkill` (charging ticks show

Bug ID	Exact Anchor	What Goes Wrong	Exact Fix
Element/Synergy/Mastery & Pokes Early		damaging poke on charge tick 1!) and `finishM: sk.M * releaseMult` — **dropping `p.synDmgMult` (`elemDmgMult * syn.dmgMult`) and `masBase`** when `resolveChannelTick` strikes!	`'⚡ กำลังรวบรวมปราณ...'` without poking!) and preserve `synDmgMult: p.synDmgMult`, `masBase`, `critAdd`, and `pen` onto the final charged release strike!
8C. `slotSig` vs `applyEquipmentSkillModifiers` Key Mismatch	`skills.js:301` vs `martial.js:527`	`SK.list`'s cache signature `slotSig` reads `m.targetTechId || m.targetArch`, whereas `applyEquipmentSkillModifiers` actually reads `mod.matchTechId`, `mod.matchArch || mod.arch`, and `mod.matchElement || mod.element`!	Align `slotSig` to hash `(m.matchTechId || m.targetTechId)`, `(m.matchArch || m.arch || m.targetArch)`, `(m.matchElement || m.element)`, and `m.makeAoe`.
8D. `SHADOW_SAMURAI` (`qiMax = 0`) Casts High-Cost Ultimates for Free	`skills.js:360` & `combat.js:803`	`SK.qiCost` multiplies `qiMax(n) * costPct / 100 = 0`, so Zero-Qi Samurai cast Tier-7 `costPct: 20%` ultimates for **0 Qi, 0 Stamina, and 0 `IAI_FOCUS`**!	For `CULT.isZeroQiSamurai(n)`, convert `p.costPct` into a bounded **Stamina (`energy`) + `IAI_FOCUS` / Tempo cost** (`staminaCost = clamp(p.costPct * 0.22, 0.5, 4.5)`), requiring `energy >=`

Bug ID	Exact Anchor	What Goes Wrong	Exact Fix
			`staminaCost + 6` in `chooseSkill`!

---

9. Mandatory Verification Suite Spec (For System #03 Part 1)

Add `test/v270_martial_moves_part1_suite.js` asserting:

1. **`test_all_54_art190_mantra_tagged_techniques_produce_valid_combat_profiles`**: Assert `SK.profile(CULT.tech('ART190_XIAN_002')) !== null`, `SK.profile(CULT.tech('ART190_BODY_002')) !== null`, and `SK.profile(CULT.tech('ART190_GOD_002')) !== null`.
2. **`test_sk_list_cache_unlocks_multi_move_on_mastery_gain_without_stages_array`**: Give an NPC a `moveDefs`-only multi-move technique; call `SK.list(n)` (1 move), then `SK.gainMastery(W, n, techId, 30)`, and assert `SK.list(n).length >= 2` immediately without needing equipment swaps.
3. **`test_choose_skill_casts_tier1_starters_and_scales_control_channel_by_tier`**: Verify a Stage-1 `XIAN` Sword Immortal casts `ART190_XIAN_001` Move 1 over basic attack, and prefers Tier-6 `M5_DUGU_MANTRA` over a Tier-1 `control` move.
4. **`test_practice_and_spar_grant_technique_mastery_and_unlock_moves`**: Run `A.practice` + `C.sparRound` and assert `n.cult.tm[techId] > 0` and real move names appear in `a.fx.text`.
5. **`test_16_weapon_classes_match_named_techniques_and_dynamic_stance_range`**: Equip a `FAN` with `ART190_XIAN_007` and `ZITHER` with `ART190_XIAN_008`; assert `M.weaponCompatibility(n, t).state === 'MATCHED'` (`1.15x`, not `0.65x`), and verify `THUNDER_STEP_STANCE` increases live `resolveMoveSpatialSpec(p, n).max` by `+0.55`.